Abstract

This thesis is a series of publications that introduce novel methods for human neural rendering using limited information, focusing on Neural Radiance Fields (NeRFs) and 3D Gaussian Splatting (3DGS). It explores how these models construct 3D representations from 2D images and demonstrates ways to condition these representations for generating high-quality human renderings. We propose techniques that use simple, interpretable inputs derived from sparse training data and extends these methods to perform effectively in few-shot learning scenarios.

We begin by examining the field of neural radiance fields, addressing limitations in existing approaches and presenting contributions to controllable radiance fields. By incorporating partial and sparse data during training, it leverages the smoothness of neural networks to produce controllable, high-quality human images.

To tackle the reliance on extensive, high-quality data annotations from multi-view videos, we introduce a new method for training neural radiance fields in few-shot, multi-view settings. This approach learns internal deformation templates, which blend smoothly during inference, significantly improving image quality compared to existing baselines and enabling effective human rendering from limited input images.

The work also addresses the need for adaptable computational efficiency during inference. It proposes a fine-to-coarse learning strategy for 3D Gaussian Splatting, which upscales a latent 2D grid that stores Gaussian representations. This strategy achieves competitive results while allowing deployment on various computational devices with minimal quality loss.

In addition, we develop a novel model for controlling radiance fields through environmental lighting. By incorporating precomputed radiance transfer, this model enables physically plausible scene relighting and provides users with intuitive control over lighting in reconstructed scenes.

This research advances the state of the art in controllable neural radiance fields and expands their application to few-shot learning scenarios. These innovations enhance the possibilities for human rendering from limited information and open new directions for future research in the field.

Keywords: Neural Rendering, Neural Radiance Fields, Few-Shot Learning, Human Rendering, Partial Information, Gaussian Splatting

Publications in this thesis

Title	Authors	Venue	Status
CoNeRF: Controllable Neural Radiance Fields	**Kacper Kania**, Kwang Moo Yi, Marek Kowalski, Tomasz Trzciński, Andrea Tagliasacchi	CVPR 2022	Accepted
BlendFields: Few-Shot Example-Driven Facial Modeling	**Kacper Kania**, Stephan J. Garbin, Andrea Taglisacchi, Virginia Estellers, Kwang Moo Yi, Julien Valentin, Tomasz Trzciński, Marek Kowalski	CVPR 2023	Accepted
LumiGauss: High-Fidelity Outdoor Relighting with 2D Gaussian Splatting	Joanna Kaleta, **Kacper Kania**, Tomasz Trzciński, Marek Kowalski	WACV 2025	Accepted
CLoG: Leveraging UV Space for Continuous Levels of Detail	**Kacper Kania**, Rawal Khirodkar, Shunsuke Saito, Kwang Moo Yi, Julieta Martinez	CVPR 2025	Under Review